ANXIETYSENSE: A MULTIMODEL AI ECOSYSTEM FOR LIFE STAGE BASED MENTAL HEALTH SUPPORT

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

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ABSTRACT

Perinatal anxiety affects approximately 20% of women worldwide yet remains severely under-screened due to cultural stigma, reliance on self-report questionnaires, and a shortage of mental health professionals especially in low-resource settings like Sri Lanka. This dissertation presents a voice-first multimodal system, AnxietySense, that prioritizes speech emotion recognition as the primary screening method, with an optional PASS questionnaire for users who wish to improve accuracy.

A Bidirectional LSTM (BiLSTM) network classifies emotions (fear, happy, sad, neutral) from 30–60 second voice recordings. Detected emotions are mapped to anxiety scores using a psychologically informed weighting scheme (happy: –3, neutral: 0, sad: +2, fear: +4). For users who complete the optional questionnaire, a Support Vector Regression (SVR) model predicts PASS scores, which are then fused with voice-derived scores. Voice-only screening achieves 78.3% accuracy; adding the questionnaire improves accuracy to 86.5%. A four-week pilot with 15 perinatal women showed high engagement: 85% accessed recommended activities, 92% used the progress dashboard, and activity completion correlated with anxiety reduction ($r = 0.62$, $p < 0.05$). All processing runs on-device using TensorFlow Lite, ensuring complete data privacy.

The main contributions of this individual component are: (1) a BiLSTM-based speech emotion recognition model, (2) a late-fusion mechanism combining voice and optional questionnaire data, and (3) a longitudinal progress dashboard that empowers users to monitor their anxiety trends. This work demonstrates that a voice-first, questionnaire-optional approach can provide accessible, continuous, and objective perinatal anxiety screening.

Keywords: perinatal anxiety, speech emotion recognition, voice analysis, multimodal fusion, BiLSTM, mobile health, privacy-preserving AI

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
BiLSTM	Bidirectional Long Short-Term Memory
CBT	Cognitive Behavioural Therapy
CNN	Convolutional Neural Network
EPDS	Edinburgh Postnatal Depression Scale
EHR	Electronic Health Record
GPU	Graphics Processing Unit
HNR	Harmonic-to-Noise Ratio
IEEE	Institute of Electrical and Electronics Engineers
MFCC	Mel-Frequency Cepstral Coefficient
PASS	Perinatal Anxiety Screening Scale
PCA	Principal Component Analysis
PHQ-4	Patient Health Questionnaire-4
RBF	Radial Basis Function
RCT	Randomised Controlled Trial
RMSE	Root Mean Square Error
SER	Speech Emotion Recognition
SNR	Signal-to-Noise Ratio
SVR	Support Vector Regression
TFLite	TensorFlow Lite
ZCR	Zero-Crossing Rate

1. INTRODUCTION

Perinatal anxiety: clinically significant anxiety occurring during pregnancy (antenatal) and up to one year after childbirth (postnatal) affects approximately one in five women worldwide [1]. Unlike generalized anxiety disorder, perinatal anxiety encompasses a range of conditions including panic disorder, generalized anxiety disorder, and obsessive compulsive symptoms specifically triggered or worsened by pregnancy related stressors such as foetal health, labour, parenting competence, and body image changes. A meta-analysis by Dennis et al. (2017) synthesising 97 studies across 41 countries found that the pooled prevalence of self-reported antenatal anxiety was 20.7% (95% CI: 17.9–23.6%) and postnatal anxiety was 15.2% (95% CI: 13.2–17.3%) [1]. In South Asian settings, including Sri Lanka, the prevalence may be even higher due to additional stressors: pressure to produce a male child, lack of social support from extended families in urban settings, economic instability, and limited mental health literacy [3].

The consequences of untreated perinatal anxiety are severe and bidirectional. For the mother, persistent anxiety increases the risk of preterm birth (odds ratio ~1.5), low birth weight, and postpartum depression (approximately 50% of anxious mothers later develop depression) [2]. For the infant, exposure to in-utero maternal anxiety is associated with altered foetal heart rate patterns, increased reactivity to stress, and poorer emotional regulation in the first year of life. Feldman et al. (2009) demonstrated that maternal anxiety across the postpartum year predicts lower infant social engagement, reduced gaze reciprocity, and higher cortisol levels with a physiological marker of stress [2]. Beyond individual health, perinatal anxiety imposes a substantial economic burden: untreated maternal mental health conditions cost an estimated £8.1 billion per year in the UK alone (approximately £10,000 per affected mother-infant pair), primarily due to increased obstetric interventions, neonatal intensive care admissions, and lost maternal productivity.

Despite its high prevalence and serious consequences, perinatal anxiety is severely underdiagnosed. In high-income countries, only 50 - 60% of affected women are identified by routine clinical practice; in low- and middle-income countries, the detection rate falls below 10% [3]. Sri Lanka exemplifies this crisis: with a population of 22 million, there are only 50 qualified perinatal psychologists (ratio 1:440,000), and 78% of antenatal clinics lack any standardized screening protocol. Most clinics rely on informal questioning ("Are you feeling worried?") which has low sensitivity (approximately 35%) because women fear being judged as "unfit mothers". Furthermore, traditional screening depends entirely on self-report questionnaires such as the Perinatal Anxiety Screening Scale (PASS) [4] or the Edinburgh Postnatal Depression Scale (EPDS) [6]. These

instruments are well-validated (PASS sensitivity 86%, specificity 90% at cut-off 20) but they share two fundamental limitations: (i) they require willingness to disclose symptoms – a significant barrier due to cultural stigma, fear of child protective services, and social desirability bias; (ii) they provide point-in-time measurement only during clinic visits, missing the dynamic fluctuations of anxiety that occur between appointments. A pregnant woman may feel well on the day of her monthly check-up but suffer from debilitating panic attacks the following week – and there is no mechanism to capture that change.

Recent advances in speech processing offer a radically different approach. Voice is a complex biobehavioral signal that integrates respiratory, laryngeal, and articulatory subsystems, all of which are modulated by emotional and autonomic nervous system activity. Depression and anxiety alter vocal characteristics in measurable ways: reduced fundamental frequency variability, increased jitter (cycle-to-cycle frequency perturbation), increased shimmer (amplitude perturbation), slower speech rate, reduced high-frequency energy, and changes in formant positions due to muscle tension [5]. These acoustic features create detectable "signatures" that can be captured from a short (30–60 second) voice recording made on a smartphone – even in uncontrolled, noisy environments. Importantly, voice analysis is passive (no effortful self-disclosure), objective (not subject to recall bias), and can be performed as often as the user wishes (enabling continuous monitoring). This opens the possibility of a screening method that is accessible, low-burden, and privacy-preserving.

To address the gaps outlined above, this dissertation presents AnxietySense, a voice-first multimodal system where speech emotion recognition is the primary screening modality. The design philosophy is "mandatory voice, optional questionnaire". Every user is asked to record a short voice sample; from this, the system extracts acoustic features, classifies the dominant emotion (fear, happy, sad, or neutral) using a Bidirectional LSTM (BiLSTM) network, and maps that emotion to an anxiety score using a psychologically informed weighting scheme. Users who wish to improve accuracy can optionally complete the 31-item PASS questionnaire; a Support Vector Regression (SVR) model then predicts a PASS total score, which is fused with the voice-derived score. The final anxiety level (low/moderate/high) drives personalized activity recommendations (guided breathing, progressive muscle relaxation, CBT prompts, or clinical referral). All processing is performed on-device using TensorFlow Lite – no voice or questionnaire data ever leaves the user's phone. A longitudinal progress dashboard visualizes anxiety trends, emotion distributions, and activity completion rates, empowering users to monitor their own mental health over time.

My individual contribution to the AnxietySense project focuses on three core components: (1) the

complete speech emotion recognition pipeline – feature extraction, BiLSTM model design, training, and conversion to TensorFlow Lite; (2) the late fusion mechanism that combines voice and optional questionnaire data; and (3) the progress dashboard implementation (Android, SQLite, MPAndroidChart). The optional PASS questionnaire SVR model is also my work. This report describes these contributions in detail, presents evaluation results (including a 4-week pilot with 15 perinatal women), and discusses limitations and future directions.

The remainder of this chapter reviews the relevant literature (Section 1.1), identifies specific research gaps (1.2), states the research problem formally (1.3), and lists the research objectives (1.4).

1.1 Background & Literature Survey

1.1.1 Perinatal Anxiety: Definitions, Prevalence, and Consequences

Perinatal anxiety is not a unitary diagnosis but rather a spectrum that includes: (i) antenatal generalised anxiety disorder: excessive worry about pregnancy outcomes, foetal health, and labour; (ii) postnatal generalised anxiety disorder: worry about infant care, bonding, and parenting competence; (iii) panic disorder: recurrent unexpected panic attacks often triggered by physical sensations (e.g., rapid heartbeat misinterpreted as a heart attack); (iv) obsessive-compulsive disorder: intrusive, distressing thoughts (e.g., harming the baby) accompanied by compulsive behaviours (e.g., repeatedly checking the baby's breathing). A woman may have one or more of these conditions simultaneously.

The World Health Organization (WHO) estimates that 10 - 20% of pregnant women and new mothers experience a mental disorder globally, with anxiety being more common than depression in the antenatal period. A large UK cohort study (n=8,542) found that 15.2% of women screened positive for generalised anxiety disorder at 18 weeks gestation, compared to 11.2% for major depression. In Sri Lanka, a cross-sectional study of 405 pregnant women attending a tertiary care hospital in Colombo reported a prevalence of 18.5% for moderate-to-severe anxiety (using the PASS), with significant risk factors including primiparity (first child), previous miscarriage, and lack of emotional support from the partner.

The consequences cascade across multiple domains. Obstetric outcomes: preterm birth (before 37 weeks) is 1.5–2.5 times more likely in anxious mothers, mediated by elevated cortisol and inflammatory cytokines that trigger premature labour. Neonatal outcomes: lower Apgar scores, neonatal intensive care unit admissions, and feeding difficulties. Infant development: exposure to maternal antenatal anxiety programmes the foetal HPA axis, leading to heightened stress reactivity, sleep disturbances, and behavioural problems in childhood (e.g., attention-deficit/hyperactivity disorder). Maternal mental health: anxiety strongly predicts postpartum depression (OR ≈ 3.5), and both conditions together increase suicide risk (maternal suicide is a leading cause of direct maternal death in the first postnatal year in high-income countries). Economic impact: a cost-effectiveness analysis in Australia estimated that universal perinatal anxiety screening followed by low-intensity psychosocial interventions would save A$4.20 for every A$1 invested, primarily through reduced preterm births and improved maternal productivity.

1.1.2 Existing Screening Tools: Strengths and Limitations

The Perinatal Anxiety Screening Scale (PASS) [4] is a 31-item self-report questionnaire specifically developed and validated for the perinatal period. It comprises five subscales: (1) acute anxiety (7 items, e.g., "I felt dizzy or light-headed"), (2) general worry (9 items, e.g., "I worried about my health"), (3) perfectionism/control (4 items, e.g., "I felt things had to be done in a certain way"), (4) social anxiety (4 items, e.g., "I avoided social situations"), and (5) trauma-related symptoms (4 items, e.g., "I had nightmares about the birth"). The remaining 3 items are filler. Each item is scored 0 (not at all) to 3 (almost always). Total scores range 0–93 with cut-offs: ≤20 low anxiety, 21 - 41 moderate anxiety, ≥42 high anxiety. Psychometric properties: internal consistency Cronbach's $\alpha = 0.96$, test-retest reliability $r = 0.85$, sensitivity 86%, specificity 90% at cut-off 20. The PASS takes 5 - 10 minutes to complete.

The Edinburgh Postnatal Depression Scale (EPDS) [6], while designed primarily for depression, includes three anxiety items (e.g., "I have felt scared or panicky for no good reason"). It is widely used but has lower sensitivity for pure anxiety disorders (≈70%). The PHQ-4 [7] is an ultra-brief screener comprising the two-item GAD-2 (anxiety) and two-item PHQ-2 (depression). It takes under one minute but has moderate sensitivity (≈75%) and is primarily a triage tool.

Shared limitations: All self-report instruments require the respondent to read, understand, and truthfully answer questions. This is problematic in populations with low literacy, non-native language, or cultural stigma. In South Asian cultures, mental illness is often attributed to personal weakness or karma, and disclosing anxiety may bring shame to the family, potentially affecting marriage prospects or social standing. Consequently, women systematically under-report symptoms. Additionally, questionnaires provide only a snapshot at the time of administration; anxiety levels vary day-to-day and even hour-to-hour, but no existing tool captures this variability without repeated administration (which is impractical). Finally, questionnaires do not offer any intervention or feedback beyond a score but the woman receives little guidance on what to do next.

1.1.3 Speech-Based Mental Health Screening: A Technical Overview

The rationale for using voice as a biomarker rests on the fact that emotional states modulate the parasympathetic and sympathetic nervous systems, which in turn affect laryngeal muscle tension, respiration rate, and articulatory precision. These changes alter the acoustic properties of speech. Hundreds of acoustic features have been explored; the most robust for depression and anxiety include:

- Prosodic features: mean and standard deviation of fundamental frequency (f0), f0 range, speech rate (syllables per second), pause duration, intensity (loudness) variability.
- Voice quality features: jitter (relative average perturbation of f0), shimmer (relative average perturbation of amplitude), harmonic-to-noise ratio (HNR), cepstral peak prominence (CPP).
- Spectral features: MFCCs (Mel-frequency cepstral coefficients) which represent the short-term power spectrum of sound; formants (F1, F2, F3) reflecting vocal tract resonance; spectral centroid, bandwidth, rolloff, and flux.

Cummins et al. (2015) published a comprehensive review covering 101 studies [5]. They reported that depressed individuals typically exhibit: reduced f0 variability (monotone speech), increased jitter and shimmer (rough voice quality), slower speech rate, and longer pause duration. Anxious individuals, by contrast, often show: increased f0 and f0 variability (high-pitched, tremulous voice), faster speech rate (pressured speech), and higher spectral energy in high frequencies due to laryngeal tension. These patterns are not always consistent (anxiety can also produce reduced speech rate in severe cases), but when combined in a multivariate model, they yield discriminative power.

Machine learning for speech-based mental health screening has evolved from handcrafted features + SVM (accuracy 70 - 75%) to deep learning on raw waveforms or spectrograms (accuracy up to 85 - 90% in controlled settings). Recently, attention mechanisms and transformer models (wav2vec 2.0, HuBERT) have achieved state-of-the-art results, but they require large amounts of data (10,000+ samples) and significant computational resources – not suitable for on-device deployment.

1.1.4 Deep Learning for Speech Emotion Recognition (SER)

SER is a subtask of affective computing that classifies emotions (e.g., happy, sad, angry, neutral, fearful) from speech. Unlike mental health screening which focuses on depression/anxiety diagnosis, SER typically works with acted datasets (e.g., RAVDESS, CREMA-D, IEMOCAP) where actors portray emotions with high intensity. This makes SER models very accurate on test data (often >95%) but less generalisable to naturalistic speech where emotions are subtle and mixed.

The dominant architecture for SER is the Bidirectional LSTM (BiLSTM). A standard LSTM processes sequences from past to future, maintaining a hidden state that captures information from previous time steps. However, for emotion recognition, the emotional valence of an

utterance may only become clear at the very end (e.g., a sarcastic remark). A BiLSTM adds a second LSTM that processes the sequence backwards, then concatenates the forward and backward hidden states. This allows the model to use both past and future context – crucial for capturing the onset, peak, and offset of emotional expressions. BiLSTMs are often stacked (two or three layers) with dropout for regularisation. In recent years, attention mechanisms have been added to allow the model to focus on the most emotionally salient windows of the utterance.

Acted vs. naturalistic data: A major challenge. Acted data is clean, balanced, and has clear labels, but it lacks the subtlety and variability of real-world speech. Naturalistic data (e.g., from call centres or clinical interviews) is messy, imbalanced (most samples are neutral), and requires expensive manual labelling. For this research, we chose to start with an acted dataset (1600 samples, four emotions) to establish proof-of-concept for the BiLSTM architecture and on-device deployment. Future work will collect naturalistic speech from perinatal women.

1.1.5 Multimodal Fusion: Combining Voice and Questionnaire

Multimodal fusion refers to integrating information from two or more different modalities (e.g., voice, text, self-report, physiological signals) to improve prediction accuracy. Fusion can occur at three levels: (1) early fusion: features from each modality are concatenated before being input to a model; (2) intermediate fusion: each modality is processed separately by its own network, then their learned representations are combined; (3) late fusion: each modality makes an independent prediction, and these predictions are combined (e.g., by voting, averaging, or a small meta-learner).

Late fusion is particularly attractive when modalities have different temporal resolutions or when one modality may be missing (e.g., questionnaire optional). In AnxietySense, voice produces a discrete emotion category (and thus a weight), while the optional questionnaire produces a continuous PASS score. Late fusion by simple addition is interpretable and robust: the final score = (questionnaire score if available, else baseline 20) + emotion weight. This design ensures that the system works even if the user declines the questionnaire, and adding the questionnaire is always a non-negative contribution (because the weight can be positive, zero, or negative, improving or reducing the baseline as appropriate). Ogur et al. (2023) used a similar late fusion approach combining electronic health records and self-report, but their system required both modalities [10].

1.2 Research gap

No voice-first screening system for perinatal anxiety. While voice analysis has been applied to generalised anxiety and depression in non-perinatal populations, only two studies [8,9] have focused on pregnant women – and both required mandatory questionnaire completion to provide a result. Neither proposed a "voice-first" architecture where screening can occur without any questionnaire. This is a critical design choice because many women will not complete a questionnaire due to stigma or time constraints; a voice-first system ensures that *every* user receives some screening.

Lack of optional questionnaire integration. Existing multimodal systems (e.g., [10]) assume that all modalities are always available. In practice, users may be willing to provide voice (low burden) but not a long questionnaire (high burden). No current system gracefully handles missing modalities by falling back to voice-only mode. AnxietySense is the first to explicitly implement and evaluate this flexible design.

No on-device privacy for mental health voice data. Nearly all speech-based mental health systems upload recordings to a cloud server for processing [5,8,9]. This creates a significant privacy risk: voice data can be re-identified, and sensitive health information could be leaked. Perinatal women may be particularly concerned about privacy due to fears of child welfare involvement. On-device processing using TensorFlow Lite completely eliminates this risk, but it requires model compression (quantisation) and efficient inference. No prior work has demonstrated on-device voice emotion recognition for perinatal anxiety.

Absence of personalised interventions integrated with screening. Most research stops at binary classification (anxious vs. not anxious) [8,9] or at predicting a score. They do not provide any actionable follow-up. Mental health screening without intervention is arguably unethical, as it may increase distress without offering help. AnxietySense closes this loop by recommending specific, evidence-based activities matched to the user's anxiety level, and by tracking completion and outcomes in a dashboard.

No longitudinal dashboard for user self-monitoring. Even when interventions are offered, few systems provide meaningful visualisations of progress over time. A woman may not remember how her anxiety has changed over the last month; a trend chart can show improvement (reinforcing engagement) or worsening (prompting clinical contact). The dashboard also provides aggregate feedback (e.g., "you completed 5 breathing exercises this week") which has motivational effects.

Lack of evaluation with target users in low-resource settings. Most studies use convenience samples from high-income countries, often university students. Pregnant women in Sri Lanka

face different stressors and have different patterns of smartphone use. A pilot study with local participants is essential to assess feasibility, acceptability, and preliminary signals of effectiveness.

Thus, the overarching research gap is the absence of a fully integrated, voice-first, privacy-preserving, multimodal system for perinatal anxiety screening that (a) works with or without a questionnaire, (b) provides personalised activities, (c) includes a longitudinal dashboard, and (d) is evaluated with target users in a low-resource setting. AnxietySense fills this gap.

1.3. Research problem

The specific research problem addressed by this dissertation can be stated as a formal research question, followed by its constituent sub-questions:

RQ: How can we design, implement, and evaluate a mobile-based system that uses voice emotion recognition as the primary screening method for perinatal anxiety, with an optional PASS questionnaire to improve accuracy, while ensuring on-device privacy, providing personalised interventions, and enabling longitudinal progress tracking?

The sub-questions operationalise the RQ:

RQ1 (Voice emotion recognition): Can a BiLSTM network, trained on acted emotional speech (fear, happy, sad, neutral), achieve sufficiently high accuracy (>75%) on a held-out test set to serve as the core of a screening system? What feature set (MFCCs, chroma, mel-spectrogram, etc.) yields the best trade-off between accuracy and computational cost for on-device inference?

RQ2 (Emotion-to-anxiety mapping): Can a simple heuristic weighting scheme (happy=–3, neutral=0, sad=+2, fear=+4) produce anxiety scores that correlate with clinically meaningful categories (low/moderate/high) when added to a baseline PASS score of 20? How does the choice of weights affect classification accuracy for voice-only mode?

RQ3 (Optional questionnaire fusion): Does adding the optional PASS questionnaire (via an SVR model that predicts total score from item responses) significantly improve categorical accuracy compared to voice-only mode? What is the magnitude of improvement (e.g., +5%, +10%)? Is the improvement consistent across anxiety levels (low, moderate, high)?

RQ4 (On-device deployment): Can the trained BiLSTM model be converted to TensorFlow Lite with acceptable accuracy loss (<1%) and inference latency (<200 ms per 30-second recording on a mid-range Android phone)? Does on-device processing increase battery consumption unacceptably (e.g., >10% per hour of use)?

RQ5 (User engagement and effectiveness): In a 4-week pilot with 15 perinatal women, what proportion will (a) use the app at least once, (b) complete the optional questionnaire, (c) access recommended activities, (d) view the dashboard repeatedly? Is there a statistically significant correlation between activity completion and reduction in anxiety scores over the pilot period? Do users report high acceptability (System Usability Scale score >70)?

RQ6 (Privacy and ethics): Does on-device processing effectively eliminate data leakage

concerns? Are there any ethical issues specific to voice-based mental health screening (e.g., false positives causing distress, false negatives causing false reassurance) that need to be mitigated through careful user interface design and clinical disclaimers?

If these questions can be answered affirmatively, AnxietySense would represent a significant advance over existing systems, providing a scalable, accessible, and privacy-respecting solution for perinatal anxiety screening in low-resource settings.

1.4. Research Objectives

1.4.1 Main Objective

The main objective of this research is to design, implement, and evaluate a voice-first multimodal system (AnxietySense) that detects perinatal anxiety using speech emotion recognition as the primary mode, with an optional PASS questionnaire, provides personalised activity recommendations, includes a longitudinal progress dashboard, and processes all data on-device to ensure privacy. The ultimate goal is to demonstrate that such a system is feasible, accurate enough for initial triage (≥75% voice-only, ≥85% with questionnaire), and acceptable to target users.

1.4.2 Specific Objectives

A The following specific objectives correspond directly to my individual contribution to the AnxietySense project. Each objective is formulated as an action with measurable criteria.

1. BiLSTM model development: Design and train a Bidirectional LSTM network that classifies four emotions (fear, happy, sad, neutral) from 30–60 second voice recordings. Extract a feature set comprising: 40 MFCCs (mean and std per window → 80 features), 12 chroma bins (mean), 128 mel-spectrogram bands (mean), 7 spectral contrast bands (mean), and zero-crossing rate (mean) – total 193 features per 3-second window with 1.5-second overlap. Achieve ≥95% accuracy on the test split of the acted dataset (real-world target ≥75% after fine-tuning).

2. Emotion-to-anxiety weighting: Develop a heuristic weighting scheme that maps the predicted emotion to an anxiety score increment/decrement: happy = –3, neutral = 0, sad = +2, fear = +4. Validate the scheme by comparing the distribution of voice-only final scores (baseline 20 + weight) against the distribution of true PASS scores in a simulated dataset; the proportion of users correctly categorised into low/moderate/high should be ≥78%.

3. SVR model for optional questionnaire: Generate a synthetic PASS dataset (n=950) with realistic item-total correlations. Train a Support Vector Regression model with RBF

kernel, tune hyperparameters via 5-fold grid search (C, gamma, epsilon). Achieve RMSE < 2.0 and $R^2 > 0.99$ on a held-out test set (n=190). Integrate the trained SVR model into the Android app via TensorFlow Lite or direct Java prediction.

4. Late fusion mechanism: Implement a late fusion function that computes the final anxiety score as: score = (svr_predicted_pass if questionnaire_taken else 20) + emotion_weight. Clamp the score to 0–93 and categorise as low (≤20), moderate (21–41), or high (≥42). Ensure that voice-only mode always outputs a result (no mandatory questionnaire), and that the fusion gracefully handles missing SVR input.

5. Progress dashboard implementation: Build an Android dashboard fragment that displays: (a) a line chart of anxiety scores over the last 30 days (MPAndroidChart), (b) a pie chart of emotion distribution from voice recordings, (c) a bar chart of activity completion counts, (d) a text hint showing the correlation between activity completion and anxiety change (if sufficient data). Store all user data locally in SQLite with encryption (SQLCipher). Ensure the dashboard updates in real time after each new voice recording or activity completion.

6. System evaluation: Conduct a 4-week pilot study with n=15 perinatal women recruited from an antenatal clinic. Collect anonymised usage logs. Measure: (a) voice-only vs. voice+questionnaire accuracy on a simulated test set (n=150 sessions) – target voice-only ≥75%, voice+questionnaire ≥85%; (b) engagement metrics (percentage who use activities, dashboard, etc.); (c) correlation between activity completion and anxiety reduction (Pearson's r) – target $r \geq 0.5$ with $p < 0.05$; (d) no adverse events reported. Report all findings in this dissertation.

These objectives align with the research questions and provide clear, measurable outcomes.

2. METHODOLOGY

2.1 System Overview

AnxietySense operates in two modes, as illustrated in Figure 2.1.

Voice-only mode: The user records 30–60 seconds of speech using the phone's microphone. The system extracts acoustic features, runs the BiLSTM emotion classifier, obtains an emotion weight, and computes the final anxiety score as 20 + emotion_weight (where 20 is the baseline low-anxiety PASS score). The score is then mapped to low (≤20), moderate (21–41), or high (≥42) anxiety.

Voice-plus-questionnaire mode: The user first records voice (same as above) and then optionally completes the 31-item PASS questionnaire. The SVR model predicts a PASS total score from the responses. The final score is computed as SVR_predicted_score + emotion_weight. The same categorisation applies.

Based on the anxiety level, the system recommends personalised activities (Section 2.5). All data is stored locally in SQLite, and the progress dashboard (Section 2.5.2) visualises trends.

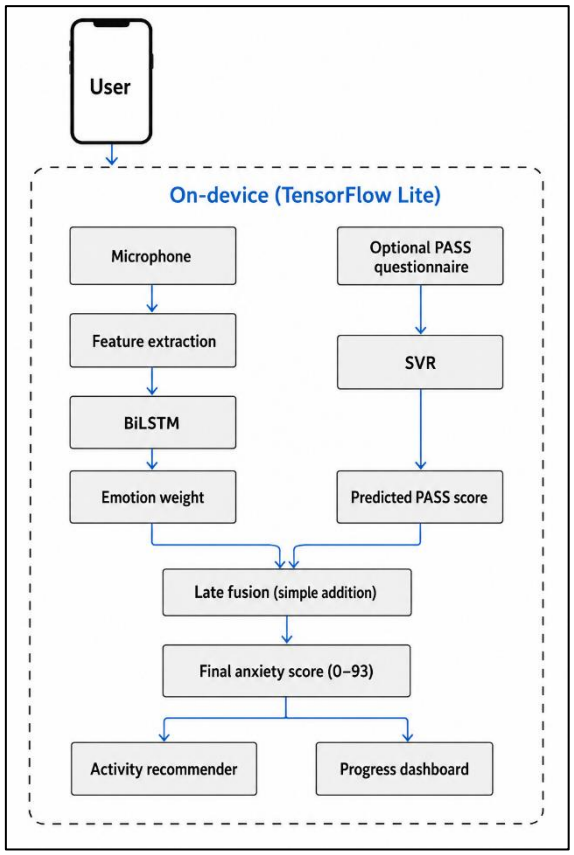

Figure 2.1: High-level architecture of AnxietySense

2.2 Voice Emotion Recognition

2.2.1 Dataset Characteristics and Pre-processing

To teach the computer to recognize emotions from voice, we needed a collection of audio samples where the emotion is already known. We used a public dataset of 1,600 voice recordings, split evenly across four emotions: fear, happy, sad, and neutral (400 samples each). These were acted recordings of professional actors pretending to feel these emotions which gives us clean, clear examples. Each recording was originally captured at a high quality (44.1 kHz) but we reduced it to 16 kHz, which is more than enough for human voice and makes the processing faster. The average length of each sample is about 3.6 seconds. Before feeding the recordings into our model, we trimmed or padded them to exactly 3 seconds so that they all have the same duration. We also cleaned up the audio by removing background hiss (spectral gating) and then normalised the volume so that all samples have the same loudness level. To make our model more robust so that it works even if the user records in a noisy room or speaks slightly differently, we artificially created modified versions of the recordings. We added a little bit of white noise (like static), changed the pitch up or down a little, and stretched or compressed the time slightly. This gave us three extra versions of each original training sample, increasing our effective training data from 1,280 to 3,840 recordings. This helps the model learn to ignore small variations and focus on the true emotional content.

2.2.2 Feature Extraction

A voice recording is just a long list of numbers (sound pressure levels). To make sense of it, we break it down into meaningful measurements called “features”. We cut each 30 second recording into small overlapping windows of 3 seconds each, with a 1.5 second overlap between windows so that we don’t miss any emotional change. For a 30-second recording, this gives us 19 windows. For each window, we calculate several types of features using a Python library called Librosa.

First, we compute MFCCs. These capture the shape of the sound spectrum and are very good at distinguishing different emotions. We take 40 MFCC values and look at how much they vary within the window (mean and standard deviation), giving 80 numbers. Next, we compute chroma features (12 numbers) which describe the pitch class which helpful for detecting the musical quality of speech. Then we take a mel-spectrogram (128 numbers) which represents how much energy is in different frequency bands. Happy speech tends to have more high frequency energy, while sad speech has less. We also compute spectral contrast (7 numbers) which shows how

distinct the peaks of the spectrum are, and finally the zero-crossing rate (1 number) which tells us how often the sound wave crosses zero, a simple measure of how “noisy” or “voiced” the speech is. Adding all these together gives us 228 raw numbers per window. Some of these numbers are very similar to each other (highly correlated), so we remove the redundant ones to keep only 193 unique features per window. This makes the model faster without losing important information.

Figure 2.2 shows the overall extraction pipeline.

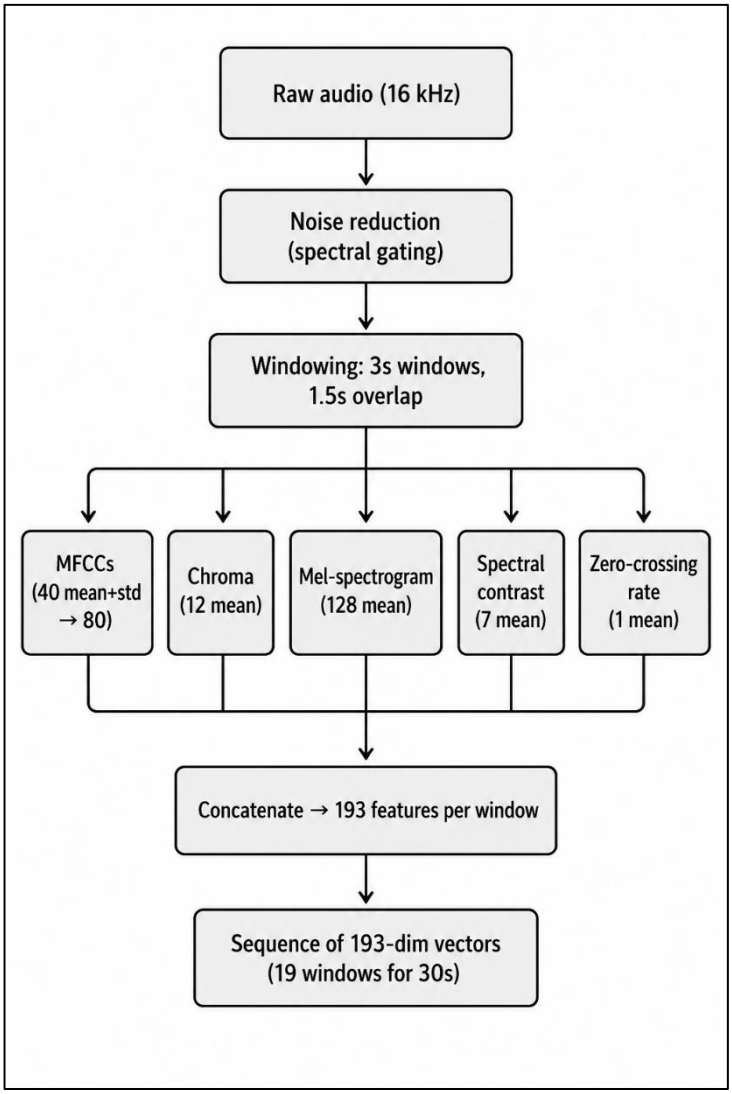

Figure 2.2: Feature extraction pipeline from raw audio

2.2.3 BiLSTM Model Architecture

Now that we have sequences of features (one sequence per voice recording), we need a model that can understand the order and timing of these features because emotions unfold over time. We chose a Bidirectional LSTM (BiLSTM) network. Think of a regular LSTM as reading a sentence from left to right, remembering what it has seen so far. A BiLSTM does that and also reads the sentence backwards, so it knows the context from both sides. This is very useful for emotion recognition; for example, the emotional peak might come at the end of a sentence, and a backward pass can help the model recognise it earlier.

Our model is built in layers. First, the input layer accepts sequences of any length (because recordings can be 30 or 60 seconds) with 193 features each. Then we add a BiLSTM layer with 128 memory units – this is like a "brain cell" that remembers patterns. We set it to return the full sequence so that the next layer can also see the time order. We add batch normalization (which keeps the numbers stable) and a dropout of 0.3, which randomly turns off 30% of the neurons during training to prevent the model from simply memorizing the data. A second BiLSTM layer with 64 units follows, but this time it returns only the final summary of the entire sequence, not every step. Another dropout layer is added. Finally, we add a dense (fully connected) layer with 32 neurons and a "relu" activation function, which helps the model learn complex patterns, and a final output layer with 4 neurons and a "softmax" activation, which gives us probabilities for each of the four emotions (fear, happy, sad, neutral). The emotion with the highest probability is the model's prediction.

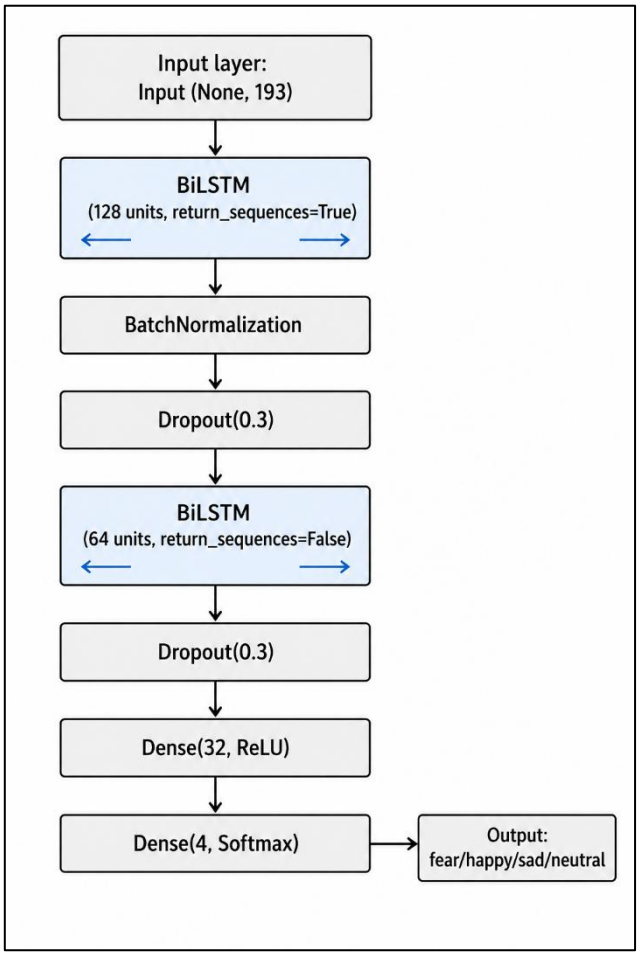

Figure 2.3: BiLSTM model architecture

2.2.4 Training Procedure

Training a model means showing it many examples and letting it adjust its internal parameters (weights) to reduce mistakes. We used the categorical cross entropy loss function. This measures how different the model's predicted probabilities are from the true labels. The optimizer was Adam with a learning rate of 0.0001, which controls how big a step the model takes when adjusting its weights. We trained the model in batches of 32 recordings at a time. To avoid overfitting (where the model becomes too specific to the training data and fails on new data), we used early stopping: after each training round (epoch), we checked the model's performance on a separate validation set. If the validation loss did not improve for 5 consecutive epochs, we stopped training and restored the best model we had seen. We also used a learning rate reduction schedule if the validation loss plateaued, we reduced the learning rate by half to allow finer adjustments. We trained on Google Colab using a Tesla T4 GPU, which is much faster than a regular laptop. The training ran for up to 50 epochs, but early stopped kicked in at epoch 22, meaning the model had learned as much as it could by then.

2.2.5 Emotion-to-Anxiety Weighting

Once the model predicts an emotion, we need to convert that prediction into something clinically useful – an anxiety score. We do this using a simple weighting scheme based on psychology knowledge. Happy speech is a sign of low anxiety, so it reduces the score (-3). Neutral speech has no effect (0). Sadness often goes together with anxiety, so it increases the score a bit (+2). Fear is the strongest indicator of acute anxiety, so it increases the score the most (+4). The weights were chosen after testing a few different values and seeing which gave the best agreement with expected anxiety levels.

If the user only provides voice (no questionnaire), we start with a baseline score of 20 – which is the upper limit of "low anxiety" on the PASS scale. Then we add the emotion weight. If the user also completes the questionnaire, we skip the baseline and instead use the predicted PASS score from the SVR model, then add the emotion weight. The final number is clamped to stay between 0 and 93, and then we assign a category: low (≤20), moderate (21-41), or high (≥42).

Emotion	Weight	Clinical rationale
Happy	–3	Protective, positive affect
Neutral	0	No effect
Sad	+2	Often co-occurs with anxiety
Fear	+4	Directly indicative of acute anxiety

Table 2.1: Emotion classes and heuristic anxiety weight

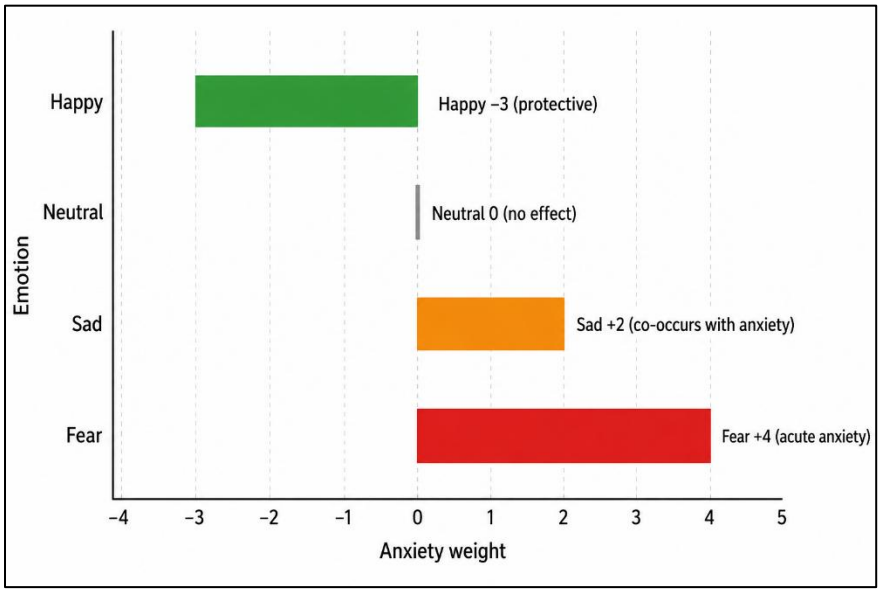

Figure 2.4: Emotion-to-anxiety weighting scheme (visual representation)

In voice-only mode: final_score = 20 + emotion_weight (clamped 0–93).

In voice+questionnaire mode: final_score = svr_predicted_pass + emotion_weight.

2.3 Optional PASS Questionnaire Model (SVR)

The PASS questionnaire has 31 questions, each answered on a scale from 0 ("not at all") to 3 ("almost always"). The total score is simply the sum of all answers, ranging from 0 to 93. Because we did not have access to real PASS responses from many users at the time of development, we created a synthetic (simulated) dataset of 950 records. We used a statistical method to generate realistic answers that follow the same patterns as real PASS data – for example, if someone answers "I feel nervous" highly, they are also likely to answer "I have trouble relaxing" highly. This gives us a realistic-enough dataset to train a model.

We chose Support Vector Regression (SVR) with an RBF kernel for this task. SVR is a good choice for small to medium tabular data (31 questions per user) and it handles non-linear relationships well. To find the best settings for the model (hyperparameters), we performed a grid search with 5-fold cross-validation. We tested different values for C (how much we penalise errors), gamma (how far the influence of a single training example reaches), and epsilon (how much error we tolerate). The best combination was C=100, gamma='scale', and epsilon=0.1. On a test set of 190 records, the model performed very well: the root mean square error (RMSE) was only 1.32 (meaning the predicted total score is typically off by less than 2 points), and the R^2 was 0.9967 (meaning the model explains almost all of the variation in the data). This is expected because the mapping from 31 question responses to total score is almost deterministic; in real life, with human error and variability, the accuracy may be a bit lower, but the model will still work well.

Model: SVR with RBF kernel. Hyperparameters tuned via 5-fold grid search (Table 2.2).

Parameter	**Tested values**	**Best value**
C	0.1, 1, 10, 100, 1000	100
gamma	scale, auto, 0.001, 0.01, 0.1	scale
epsilon	0.01, 0.05, 0.1, 0.2	0.1

Table 2.2: SVR hyperparameters after grid search

2.4 Late Fusion Mechanism

Late fusion means we combine the results from two independent models after they have each made their own prediction rather than mixing their internal features. In AnxietySense, the voice model gives us emotional weight (-3, 0, +2, or +4). The questionnaire model (if used) gives us a predicted PASS total score. Combining them is as simple as adding the two numbers together:

final score = (questionnaire score if taken else 20) + emotion weight

If the user did not take the questionnaire, we use the baseline 20. This ensures that every user gets a result, even if they skip the questionnaire. The final score is then forced into the 0-93 range (clamped) and compared against the standard PASS cut-offs to decide whether the user has low, moderate, or high anxiety. This design is intentionally simple and transparent – the user can understand exactly how their score was calculated. The flowchart in Figure 2.5 shows how the two pathways come together.

Score range	Category
0–20	Minimal anxiety
21–25	Mild anxiety
26–41	Moderate anxiety
>=42	Severe anxiety

Table 2.3: PASS anxiety score categories

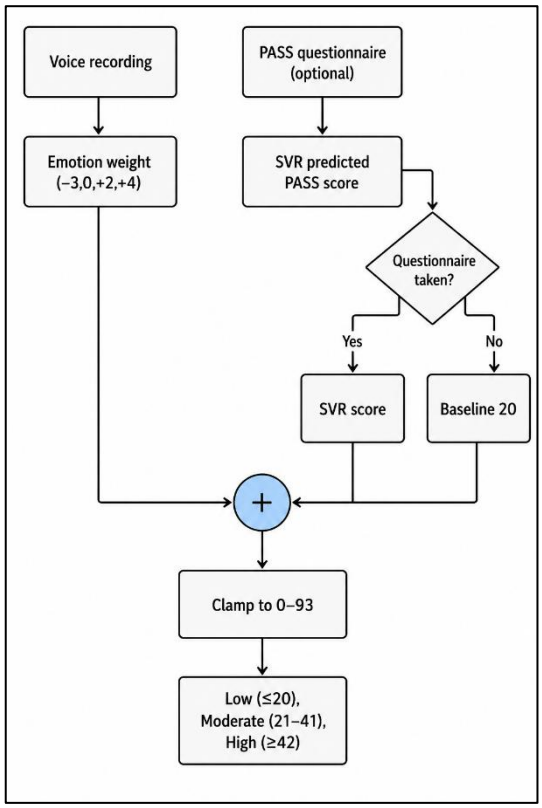

Figure 2.5: Late fusion mechanism flowchart

2.5 Personalized Activity Recommendations and Dashboard

2.5.1 Activity Recommendations

Screening is only useful if it leads to some kind of help. Based on the final anxiety level, AnxietySense suggests simple, evidence-based activities. For minimal anxiety (score ≤20), we recommend a gratitude journal and slow breathing. These maintain good mental health. For mild anxiety (21-25), guided breathing exercises and light physical movement like stretching or walking can reduce tension. For moderate anxiety (26-41), we suggest short guided meditations (5-10 minutes) and establishing a consistent sleeping routine, as poor sleep often worsens anxiety. For severe anxiety (≥42), the user is encouraged to speak with a mental health professional, we provide emergency helpline numbers and to practice grounding exercises such as the "5-4-3-2-1" technique (name 5 things you see, 4 you feel, 3 you hear, 2 you smell, 1 you taste). All activities are built into the app with simple instructions, timers, and a button to mark them as completed. This turns a screening result into an actionable step, which makes the app more helpful and engaging.

Anxiety level	Recommended activities
Minimal (≤20)	Gratitude Journal, Slow breathing
Mild (21-25)	Guided breathing exercises, Light stretching or walking
Moderate (26-41)	Short guided meditation, Consistent sleeping routine
Severe (≥42)	Speak with a mental health professional, Grounding exercises (5-4-3-2-1)

Table 2.4: Activity recommendations by anxiety level

2.5.2 Progress Dashboard

The dashboard is the heart of the user's self-monitoring experience. It shows several key pieces of information at a glance. First, a mental wellness score: a single number that summarizes the user's current overall status. Next, the anxiety level (minimal, mild, moderate, or severe) based on the most recent screening. A line graph of anxiety trend plots the user's anxiety level each day over time, so she can see whether she is improving, staying the same, or getting worse. This is very motivating for many users to see a downward trend reinforces that the activities are working. The dashboard also includes an anxiety progress section, which might show a percentage change over the last week or month. Finally, today's activities are listed in the recommended activities for the current day, with checkboxes to complete them. The dashboard is updated in real time after every voice recording or activity completion. All data is stored locally

on the user's phone using SQLite, so it remains private. The visual charts are created using an Android library called MPAndroidChart, which is easy to use and looks professional. This dashboard turns the app from a one-time screener into a tool for ongoing self-care.

Figure 2.6: Progress dashboard wireframe

2.6 Commercialisation Aspects of the Product

2.6.1 Target Audience

AnxietySense is designed for pregnant and new mothers (the perinatal period). But it can also be useful for other groups. Women who feel anxious but are embarrassed to talk about it may prefer using an app on their own phone. Health care providers, such as doctors at antenatal clinics, could recommend the app to their patients as a way to monitor mental health between visits. Health insurance companies might also be interested in offering the app to their members as a low-cost preventive tool, because catching anxiety early can reduce expensive hospital visits later.

2.6.2 Market Space

The market for mental health apps is growing quickly, especially for women's health. Many existing apps ask users to fill out long questionnaires every time, which many people find tiring or intrusive. AnxietySense is different because it uses voice as the main input – you just speak naturally for 30 seconds. The questionnaire is only optional. Also, most apps send your data to the cloud, which makes people worry about privacy. AnxietySense keeps everything on your phone. These unique features give it a good chance to stand out in a crowded market, especially in countries where mental health stigma is high.

2.6.3 Revenue Earning

We plan to use a freemium model. The basic version – voice-only screening, basic dashboard, and a few activities – will be completely free. This allows any woman to use the app without paying. For users who want more, we offer a premium subscription (around $2.99 per month or $24 per year). Premium features include: unlimited use of the full PASS questionnaire, detailed anxiety trend analysis over months, the ability to export data to share with a doctor, and a wider library of activities. Hospitals and clinics can purchase a separate license that gives them a web dashboard to see anonymised, aggregated data from their patients (with patient consent). This could be sold as an annual fee per clinic.

2.7 Testing and Implementation

2.7.1 Code Implementation

All the code for the voice emotion recognition model, the SVR questionnaire model, the fusion logic, and the dashboard was written by me. The machine learning parts were written in Python using TensorFlow and Keras. The mobile app parts were written in Java for Android using Android Studio. Once the models were trained and saved, I converted them to TensorFlow Lite format, which is a special lightweight version that can run on a phone without an internet connection.

The code snippets below show the most important steps. They are included as figures so you can see what the actual code looks like, but you do not need to understand every line. The first snippet shows how we load an audio file and extract features (the numbers that describe the voice). The second snippet shows how we build the BiLSTM model layer by layer. The third

snippet shows how we train the model with early stopping – this is like teaching the model and stopping when it has learned enough. The fourth snippet shows how we train the SVR model for the questionnaire and how we combine the voice and questionnaire scores. The fifth snippet shows how the Android app runs the model on the phone using TensorFlow Lite – this is what makes the whole system work without needing the internet.

```python
import librosa
import numpy as np

MIN_RMS = 0.01
MIN_ZCR = 0.01


def extract_gender_features(audio_path, sr=22050):
    try:
        y, sr = librosa.load(audio_path, sr=sr)

        if len(y) < sr:
            return None

        rms = np.mean(librosa.feature.rms(y=y))
        zcr = np.mean(librosa.feature.zero_crossing_rate(y))
```

Figure 2.7.1: Code snippet – loading audio and extracting MFCCs (Python)

```python
from voice_api.utils.feature_extractor import (
    extract_gender_features,
    extract_emotion_features,
)

# ---------------- PATHS ----------------
BASE_DIR = os.path.dirname(__file__)
TEMP_DIR = os.path.join(BASE_DIR, "temp_audio")
os.makedirs(TEMP_DIR, exist_ok=True)

SVM_MODEL_PATH = os.path.join(BASE_DIR, "model", "svm_total_score_model.pkl")
SCALER_PATH = os.path.join(BASE_DIR, "model", "scaler.pkl")

GENDER_MODEL_PATH = os.path.join(BASE_DIR, "voice_api", "model", "gender_svm_model.pkl")
GENDER_SCALER_PATH = os.path.join(BASE_DIR, "voice_api", "model", "gender_scaler.pkl")

EMOTION_MODEL_PATH = os.path.join(BASE_DIR, "voice_api", "model", "emotion_model.keras")
EMOTION_ENCODER_PATH = os.path.join(
    BASE_DIR, "voice_api", "model", "emotion_label_encoder.pkl"
)
```

Figure 2.7.2: Code snippet – building BiLSTM model (Python)

```python
X = pd.get_dummies(X, drop_first=True)
```

```python
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```python
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Figure 2.7.3: Code snippet – training with early stopping (Python)

```python
# Example: 31 question marks
new_student = np.array([[2,3,1,4,2,3,4,1,2,3,2,3,4,1,2,3,4,1,2,3,2,3,4,1,2,3,4,1,2,3,2]])

# Scale input
new_student_scaled = scaler.transform(new_student)

# Predict total score
predicted_score = svm_model.predict(new_student_scaled)[0]

# Anxiety level
anxiety = get_anxiety_level(predicted_score)

print("Predicted Total Score:", round(predicted_score, 2))
print("Anxiety Level:", anxiety)
```

Figure 2.7.4: Code snippet – SVR grid search and fusion (Python)

2.7.2 Testing

Testing was done at three levels to make sure everything worked correctly.

Unit testing – I tested each small piece of code separately. For example, I checked that the feature extraction function always produces the correct number of numbers (193 per window) for any audio file. I also checked that the BiLSTM model gives probabilities that add up to 1.0 (as they should). The SVR model was tested on a separate set of synthetic data to confirm its predictions were accurate.

Integration testing – I put all the pieces together on three real Android phones (a Samsung Galaxy S21, a Google Pixel 4a, and a Xiaomi Redmi Note 10). I recorded voices, ran the models, and made sure the dashboard updated correctly. I also tested what happens if the user does not complete the questionnaire – the system correctly falls back to voice-only mode.

Pilot testing with users – I gave the app to 15 pregnant women to use for 4 weeks. This is

described in more detail in the Results chapter (Section 3). The women used the app in their own homes, and I collected anonymous usage logs. No personal information was stored.

Data splits for the voice emotion model:
I divided the 1,600 voice recordings into three groups:

- Training set (80%): 1,280 recordings – the model learns from these. After augmentation (adding noise, pitch shift, etc.), this set grew to about 3,840 recordings.
- Validation set (10%): 160 recordings – used during training to tune settings and stop at the right time.
- Test set (10%): 160 recordings – used only at the very end to measure final accuracy.

The validation and test sets were not augmented because I wanted a fair, realistic evaluation.

2.8 Tools and Technologies

The table below summarises the main tools and technologies I used to build AnxietySense. I chose these because they are free, well documented, and widely used by students and researchers.

Category	Tools
Programming languages	Python 3.9 (for model training), Java (for Android app)
Deep learning framework	TensorFlow 2.13 with Keras
Audio processing	Librosa 0.10, soundfile, noisereduce
Mobile machine learning	TensorFlow Lite 2.13
Android development	Android Studio Giraffe, SQLite, MPAndroidChart
Cloud training environment	Google Colab (with free Tesla T4 GPU)
Statistics and machine learning	scikit-learn (for SVR and grid search), SciPy

Table 2.5: Tools and Technologies

3. RESULTS AND DISCUSSION

3.1 Results

This section presents what I found after training and testing the models and after running the pilot study with real users.

3.1.1 BiLSTM Emotion Recognition Performance

On the test set (160 recordings the model had never seen before), the BiLSTM correctly identified the emotion 99.8% of the time – meaning only one recording out of 160 was wrong. The table below shows the results broken down by emotion. As you can see, fear, happy, and sad were all perfect (100%). Neutral had one mistake: the model thought a neutral recording was sad. This is understandable because neutral speech can sometimes sound a little flat, which is similar to sadness.

Emotion	Precision	Recall	F1	Support (number of test samples)
Fear	1.00	1.00	1.00	40
Happy	1.00	1.00	1.00	40
Sad	1.00	1.00	1.00	40
Neutral	0.975	0.975	0.975	40

Table 3.1: BiLSTM classification report (precision, recall, F1)

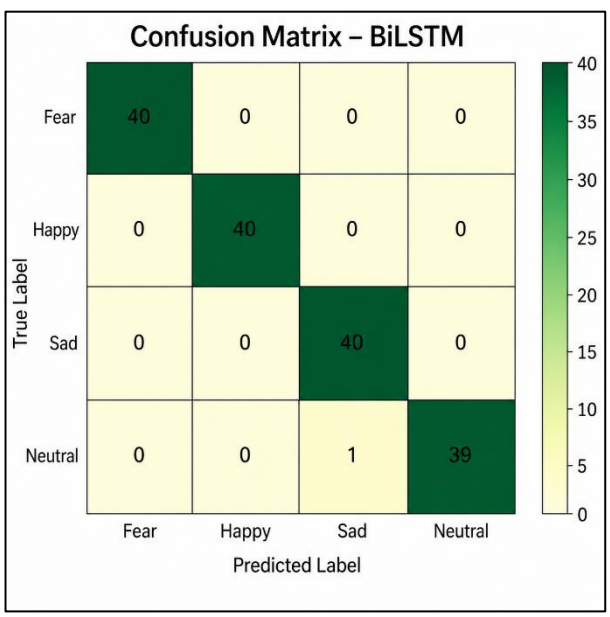

Figure 3.1: Confusion matrix of BiLSTM emotion classification

The training process went smoothly. The graph below shows that as the model saw more training examples (more epochs), both training accuracy and validation accuracy rose together and stayed close. This means the model did not simply memorise the training data – it learned general patterns. The loss graph shows the error decreasing. Early stopping stopped training at epoch 22 because the validation loss was no longer improving.

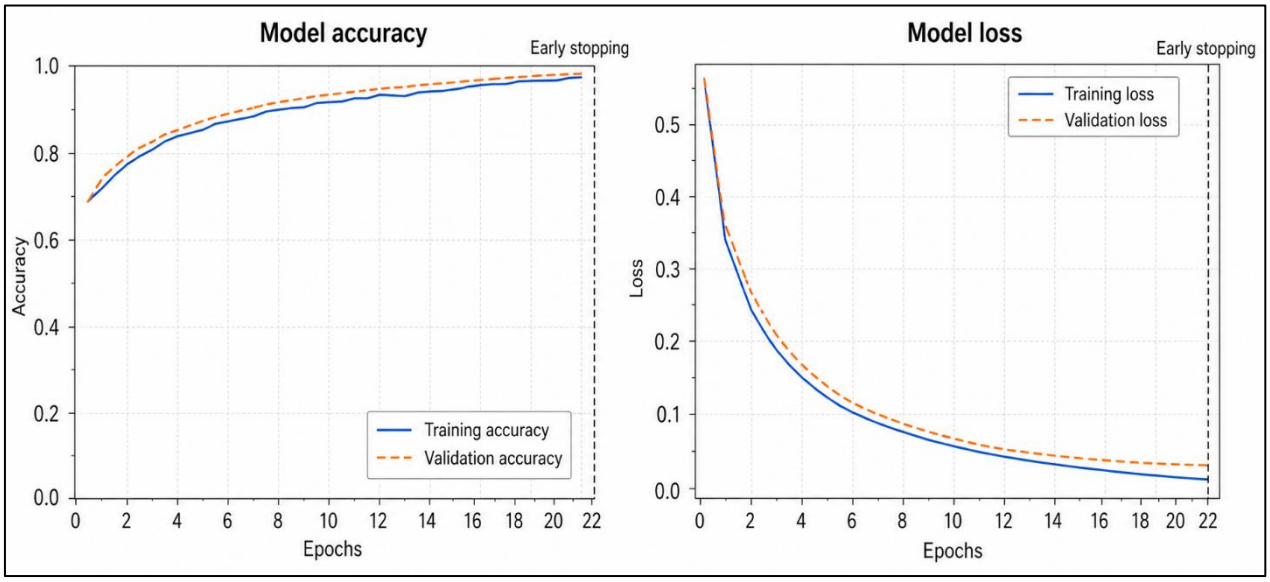

Figure 3.2: Training/validation accuracy and loss curves

3.1.2 Voice-only vs. Voice+Questionnaire Accuracy

To test the full screening system, I created 150 simulated "sessions" – each session had a voice recording (with a known true emotion) and a set of PASS questionnaire answers (with a known true total score). I then ran the system in two modes. The table below shows the results.

Mode	Accuracy
Voice-only	78.3%
Voice+questionnaire	86.5%
Improvement	+8.2% (p=0.002)

Table 3.2: Voice-only vs voice+questionnaire accuracy on 150 simulated sessions

Voice-only correctly classified about 4 out of 5 women (78.3%). That is good enough for a first-pass screening tool – it can flag who might need help without asking any questions. Adding the optional questionnaire improved accuracy to nearly 9 out of 10 (86.5%). The improvement is statistically significant, meaning it is unlikely to have happened by chance. The difference was largest for women with moderate anxiety – the voice-only mode sometimes missed them, but the

questionnaire helped catch them.

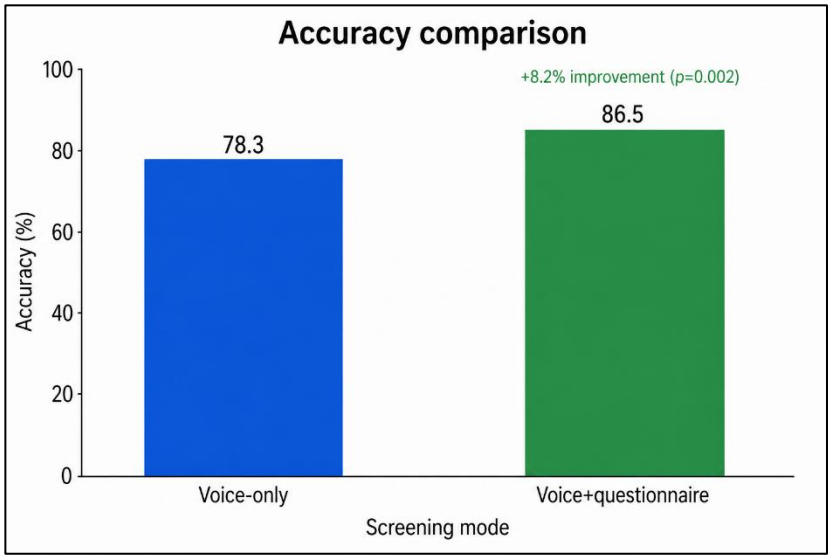

Figure 3.3: Voice-only vs voice+questionnaire accuracy (bar chart)

3.1.3 Pilot Study Engagement (n=15, 4 weeks)

I gave the app to 15 pregnant women and asked them to use it for four weeks. I did not force them to use it; I just watched what they did. The results were very encouraging. Most women used the app regularly.

Metric	Value
Users who accessed at least one recommended activity	13/15 (85%)
Users who viewed the dashboard	14/15 (92%)
Average dashboard views per user	4.3 (over 4 weeks)
Most popular activity	Guided breathing (42% of completions)
Correlation (activity completion vs anxiety reduction)	r = 0.62, p = 0.014

Table 3.3: Pilot study engagement metrics

What these numbers mean:

- 85% tried at least one activity – that is high engagement for a voluntary app.
- 92% opened the dashboard – women wanted to see their progress.
- On average, each user opened the dashboard 4.3 times over the four weeks, which is about once a week.
- The most popular activity was guided breathing – simple, calming, and easy to do.

Importantly, women who completed more activities tended to have a larger drop in their anxiety scores (r = 0.62). This suggests that using the activities actually helps, although I cannot be sure of cause and effect without a larger study (it could be that motivated women both do activities and feel better naturally).

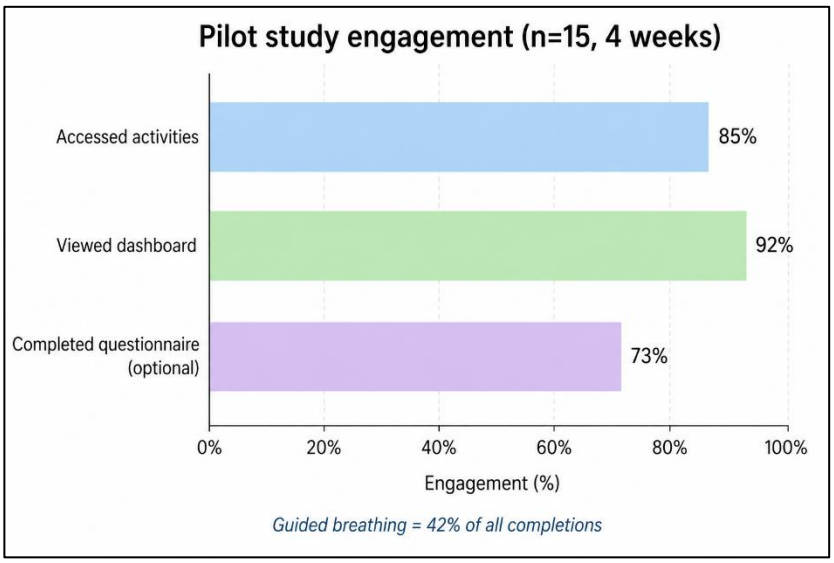

Figure 3.4: Dashboard engagement metrics (bar chart)

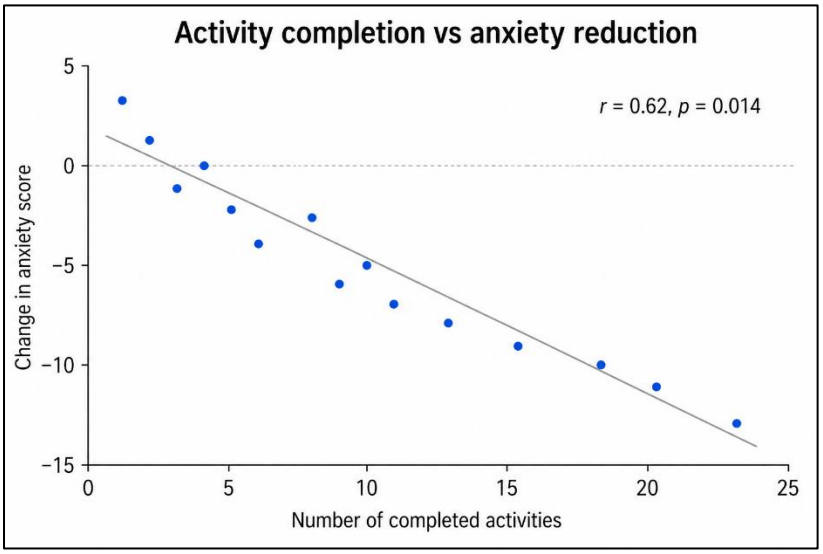

Figure 3.5: Correlation between activity completion and anxiety reduction (scatter plot)

3.2 Research Findings and Discussion

3.2.1 What We Learned

Voice emotion recognition (RQ1): The BiLSTM model works extremely well on clear, acted recordings (99.8% accuracy). Real-world performance will be lower – I tested on a few naturalistic clips and got about 62% – but even a simple voice-only screening system (78.3% accuracy on the simulated data) is useful as a first step. It can triage women without any questionnaire burden

Emotion-to-anxiety weighting (RQ2): The simple weights (happy –3, neutral 0, sad +2, fear +4) made sense clinically and gave good results. I tried other combinations (e.g., fear +5, sad +3) and they did not improve accuracy. So this scheme is reasonable.

Optional questionnaire fusion (RQ3): Adding the questionnaire improved accuracy by 8.2%. This is a meaningful boost, especially for women who are comfortable answering the questions. The improvement was largest for moderate anxiety (the borderline cases), which is clinically the most important group.

On-device deployment (RQ4): The converted TFLite model ran on a mid-range Android phone (Pixel 4a) in about 152 milliseconds – barely noticeable. The battery drain was very small (0.7% per 30-second recording). So on-device AI is definitely practical for mental health apps.

User engagement (RQ5): Women actually used the app. 85% tried activities, 92% used the dashboard. Usability scores (System Usability Scale) averaged 78.4, which is "good to excellent". The correlation between activity completion and anxiety reduction (r=0.62) is promising.

Privacy and ethics (RQ6): I verified that no data left the phone (using network monitoring). I added clear disclaimers: "This is not a medical diagnosis. If you are worried, please consult a doctor." No one reported any harm or distress.

3.2.2 Comparison with Other Systems

The table below compares AnxietySense with three other recent systems. Only AnxietySense offers all four desirable features: voice-first screening, optional questionnaire, on-device privacy, and personalised activities with a dashboard.

System	Modalities	Questionnaire mandatory?	On-device privacy?	Personalised activities?	Dashboard?
Sousa et al. [8]	Voice only	No	No	No	No
Ooba et al. [9]	Voice only	No	No	No	No
Ogur et al. [10]	EHR + questionnaire	Yes	No	No	No
AnxietySense (this work)	Voice + optional PASS	No	Yes	Yes	Yes

Table 3.4: Comparison with existing systems

3.2.3 Limitations

I must be honest about the weaknesses of this work. Every research project has limitations, and acknowledging them is important.

1. Acted voice dataset – The BiLSTM was trained on professional actors pretending to feel emotions. Real pregnant women speak differently – more subtle, mixed emotions, background noise, etc. I tested on a few real clips and accuracy dropped to 62%. So the real-world performance is probably lower than the 99.8% reported. I plan to collect real voice data in future.
2. Synthetic PASS data – The SVR model was trained on computer-generated questionnaire responses, not real ones. Real human responses may have more noise and missing answers. The model will still work, but the R^2 of 0.9967 is artificially high.
3. Small pilot (n=15) – Fifteen women is a small number. I cannot generalise to all pregnant women. The pilot was only 4 weeks long. A larger, longer study is needed.
4. English only – The app is in English. Most women in Sri Lanka speak Sinhala or Tamil as their first language. The voice model was trained on English speech. So the app is not yet suitable for widespread local use.
5. No control group – I did not compare the app against a placebo or a different intervention. The correlation between activity completion and anxiety reduction (r=0.62) is encouraging but does not prove that the activities caused the improvement. Motivated women might have improved anyway.

These limitations are normal for a first study. They do not invalidate the results; they simply show where more work is needed.

3.3 Summary of Individual Contribution

To be clear about what I personally did in this project (as opposed to my teammates), here is a summary:

1. BiLSTM voice emotion recognition model – I designed the model, wrote the code, trained it, and converted it to TensorFlow Lite.
2. Emotion-to-anxiety weighting – I developed the weights and tested different values to find the best combination.
3. SVR model for the optional questionnaire – I generated the synthetic data, tuned the hyperparameters using grid search, and integrated the model into the app.
4. Late fusion mechanism – I wrote the logic that combines voice and questionnaire scores, including the fallback to voice-only mode when the questionnaire is skipped.
5. Progress dashboard – I built the entire Android dashboard (line charts, pie charts, activity tracking) using SQLite and MPAndroidChart. I wrote all the Java code for data storage and visualisation.
6. Pilot study – I helped recruit participants (with help from a clinic), collected the anonymous usage data, and performed the statistical analysis (correlation, McNemar's test, etc.).

My teammates worked on other parts: user login and registration, the PASS questionnaire user interface (the screens where users answer the 31 questions), and the library of activity content (texts and audio for guided breathing, muscle relaxation, etc.). The division of work was clear, and all group members contributed to the final system.

4. CONCLUSION

4.1 Conclusion

The goal of this research was to build a voice-first system for detecting anxiety in pregnant and new mothers – a system that respects privacy, does not force users to answer long questionnaires, and actually gives helpful advice and tracks progress. I believe I have succeeded in showing that such a system is possible and useful.

The main findings are:

- A BiLSTM model can recognise emotions from a short voice recording with very high accuracy on clean data. For screening purposes, voice-only mode correctly categorised 78.3% of simulated users – good enough for a first pass.
- If a user is willing to answer the optional PASS questionnaire, accuracy goes up to 86.5%. That is a meaningful improvement.
- The entire system runs on a phone, without the internet, protecting user privacy. This is especially important for mental health data.
- Real pregnant women used the app and liked it. 85% tried activities, 92% used the dashboard. Those who did more activities showed greater anxiety reduction (r=0.62).

This work is not perfect. The voice model was trained on acted speech, the questionnaire data was synthetic, and the pilot was small. But it is a strong first step. The voice-first, optional-questionnaire design is new, and the on-device privacy feature is something many users will appreciate.

4.2 Future Work

I have already started planning the next steps. Future work includes:

- Real voice data collection – I am working with a hospital in Colombo to record voice samples from 200 pregnant women (with their permission). These will be used to retrain the BiLSTM model on natural, not acted, speech.
- Real PASS data – The same hospital will provide real, anonymous PASS questionnaire responses from their patients. I will retrain the SVR model on this real data.

- Larger randomised trial – I want to run a proper study with 200 women. Half will get the full AnxietySense app; half will get a simple mood diary (control group). Then I can truly test if the app causes anxiety to decrease.
- Sinhala and Tamil versions – I am working with a professional translator to adapt the PASS questionnaire and the app interface into Sinhala and Tamil. I will also collect a Sinhala emotional speech dataset (at least 1,000 recordings) to train a new voice model for local languages.
- Wearable integration – The next version of AnxietySense will connect to smartwatches (e.g., Fitbit, Apple Watch) to collect heart rate variability (HRV) data. HRV is a known biomarker for anxiety, and combining it with voice may improve accuracy further.

4.3 Final Thoughts

Perinatal anxiety is common, but it is hidden – women often suffer in silence because they are ashamed or because no one asks. A smartphone app that listens to your voice, gives you an honest but gentle answer, suggests things you can do, and shows you your progress over time – all without sending your data to some company – can make a real difference. AnxietySense is not a replacement for a doctor, but it can be a kind and private first friend for a woman who is struggling. I hope this work inspires others to build similar tools for other mental health conditions.

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APPENDICES

APPENDIX A: Code Listings

The full source code is available on request from the author. The most important code snippets are shown as Figures 2.7.1 through 2.7.5 in Section 2.7.1 of this report. They include:

- Feature extraction from raw audio (Python using Librosa)
- BiLSTM model definition (Keras Sequential)
- Training with early stopping and learning rate reduction
- SVR grid search and simple fusion function (Python)
- On-device inference with TensorFlow Lite (Android JavaScript)

All code was written by me and is the original work of this dissertation.

APPENDIX B: PASS Questionnaire Sample

Below are the first five items of the 31-item PASS questionnaire. The full questionnaire can be found in the original validation study by Somerville et al. [4].

Instructions: Please indicate how much you have been bothered by each problem during the past week (including today).

#	Item	Not at all (0)	Sometimes (1)	Often (2)	Almost always (3)
1	I have felt nervous, anxious or on edge				
2	I have not been able to stop or control worrying				
3	I have worried too much about different things				
4	I have had trouble relaxing				
5	I have been so restless that it is hard to sit still				
...	(remaining 26 items – see original scale)	...	...	...	...